

CS471 – Lesson 15 In-Class Worksheet

Reinforcement Learning: Model-Free Learning and Q-Learning

Part 1: Guided In-Class Worksheet (Instructor-Led)

Objective: Differentiate between model-based planning and model-free learning, understand experience tuples, compute temporal difference updates, and analyze exploration strategies.

1. Model-Based Planning vs. Model-Free Learning

Autonomous defense platforms operate in environments ranging from known simulations to contested battlefields.

- a. Contrast Model-Based Planning (such as Value Iteration) with Model-Free Learning (Q-learning). Identify what environmental information is missing in model-free scenarios.

- b. In model-free reinforcement learning, an agent learns entirely from experience. Define the components of an experience tuple (s, a, r, s') .

2. Step-by-Step Q-Learning Update

An autonomous loyal wingman UAV (State s) executes a “Flank” maneuver (Action a).

- Current belief: $Q(s, a) = 5.0$
- Observed immediate reward: $r = 2.0$
- Observed next state: s'
- Next state maximum expected return: $\max_{a'} Q(s', a') = 8.0$
- Discount factor: $\gamma = 0.5$
- Learning rate: $\alpha = 0.5$

- a. Calculate the **Target**: $\text{Target} = r + \gamma \max_{a'} Q(s', a')$.
- b. Calculate the **Temporal Difference (TD) Error**: $\delta = \text{Target} - Q(s, a)$. Interpret the physical meaning of the sign of δ .
- c. Compute the updated value $Q_{\text{new}}(s, a) = Q(s, a) + \alpha \delta$.

3. Exploration vs. Exploitation Tradeoff (ϵ -Greedy)

An Electronic Warfare (EW) jamming pod uses Q-learning to discover optimal frequencies to suppress hostile radars.

- a. Describe the mechanics of the ϵ -greedy action selection mechanism in terms of probabilities ϵ and $1 - \epsilon$.
- b. The EW pod operator recommends setting $\epsilon = 0$ immediately upon deployment to maximize immediate jamming effectiveness. Explain the tactical danger of this greedy strategy.

Part 2: Student Independent Practice

Objective: Apply the full Q-learning update rule, evaluate learning rate dynamics, and extract greedy policies from state-action tables.

A. Cyber-Defense Tactical Response

An automated cyber agent monitors an enterprise server (State s). Upon detecting suspicious network beaconing, it takes action a (“Isolate Port”).

- Initial Q-value: $Q(s, a) = 12.0$
 - Immediate reward received: $r = -4.0$ (due to temporary latency on critical traffic)
 - Transition state: s' (Secured Subnet)
 - Next state evaluations: $Q(s', \text{Audit}) = 10.0$, $Q(s', \text{Restore}) = 6.0$, $Q(s', \text{BlockIP}) = 14.0$
 - Discount factor: $\gamma = 0.8$
 - Learning rate: $\alpha = 0.25$
1. Identify the value of $\max_{a'} Q(s', a')$ from the candidate actions in state s' .
 2. Compute the Q-learning Target value.
 3. Compute the Temporal Difference error (δ). Did the outcome exceed or fall short of expectations?
 4. Calculate the updated value $Q(s, a)$.

B. Hyperparameter Analysis: Learning Rate and TD Error

A cyber security platform encounters an unexpected zero-day exploit and calculates a massive negative TD error of $\delta = -20.0$.

1. If the agent uses a conservative learning rate of $\alpha = 0.05$, what is the magnitude of the update adjustment?

2. If the learning rate were set to $\alpha = 0.90$, how would the agent react to this single observation? Contrast the tactical stability implications of both settings.

C. Full Update Formulation

Write out the complete, single-line mathematical formula for Q-learning. Label each of the four core conceptual steps: (1) Current belief, (2) Target, (3) TD error, and (4) Proportional adjustment.